

**STELLENBOSCH UNIVERSITY
DEPARTMENT OF ECONOMICS**

ECONOMICS 114

MAIN ASSESSMENT A2: 2 JUNE 2022

Time: 3 hours

Marks: 100

EXAMINERS:

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INTERNAL MODERATOR: Dr E Moses

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER **SECTION A** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS** NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER **SECTION B** QUESTIONS ON THE SUPPLIED **ANSWER SHEET**. INDICATE YOUR CHOICE BY MAKING A **CROSS** NEXT TO THE NUMBER OF THE STATEMENT.

KEY:

AP = Average product
AFC = Average fixed cost
AVC = Average variable cost
AC = Average total cost
BL = Budget line
GDP = Gross domestic product
IC = indifference curve
K = Capital
FF = Feasible frontier
MC = Marginal cost
MP = Marginal product
MPB = Marginal private benefit
MPC = Marginal private cost
MRS = Marginal rate of substitution

MRT = Marginal rate of transformation
MSB = Marginal social benefit
MSC = Marginal social cost
P = Price
PF = Production frontier
Q = Quantity
Q_d = Quantity demanded
Q_s = Quantity supplied
TC = Total cost
TP = Total product
w = Wage
WTA = Willingness to accept
WTP = Willingness to pay
Y = Income

SECTION A
TRUE/FALSE STATEMENTS

[10 marks: 1 mark for a correct answer and minus ½ for an incorrect answer]

1. In a game with two players, where Player 1 has three possible strategies from which she can choose and Player 2 has three possible strategies from which he can choose, there are nine possible outcomes.
 - a) True
 - b) False

2. The game shown below is an example of the prisoner's dilemma game.

		Sam	
		Wear mask	Don't wear mask
Jemima	Wear mask	5, 5	2, 6
	Don't wear mask	6, 2	3, 3

- a) True
 - b) False
3. Chani is a landowner and Paul is a spice farmer who pays a share of his output to her for the use of her land. New technology that increases Paul's productivity would increase the area of the economically feasible set.
 - a) True
 - b) False
4. An allocation will be substantively fair for all participants in a strategic interaction if it is Pareto efficient.
 - a) True
 - b) False
5. When workers are paid their reservation wage in the labour discipline model, employers have no power over them.
 - a) True
 - b) False
6. The principal-agent problem of ownership versus control of the firm arises when owners and managers pursue objectives that differ from those that the government wishes them to pursue.
 - a) True
 - b) False

7. The demand for exported Russian natural gas is price inelastic. If Russia decreases its supply of exported gas by 5 percent, then the price of gas will increase by more than 5 percent.
- a) True
 - b) False
8. There are two sellers of soft serve ice-cream in Funtown. One day, ice-cream seller A decides to enhance the quality of her product by adding chocolate sprinkles to it. Initially, this lures some customers away from ice-cream seller B's kiosk, but B soon manages to get all the lost customers back by offering a price reduction. In the end, B's quantity of ice-creams sold are the same as before. From these changes, we can conclude that the demand curve for B's ice-cream first shifted leftwards and then shifted rightwards.
- a) True
 - b) False
9. In an economy where firms are profit maximisers and markets are competitive (no government intervention), if there are negative externalities then there will be overproduction and $MSB > MSC$.
- a) True
 - b) False
10. When markets allocate resources in a Pareto-inefficient way, this is known as market failure.
- a) True
 - b) False

SECTION B
MULTIPLE CHOICE QUESTIONS

[30 marks: 2 marks for a correct answer and minus ¼ for an incorrect answer]

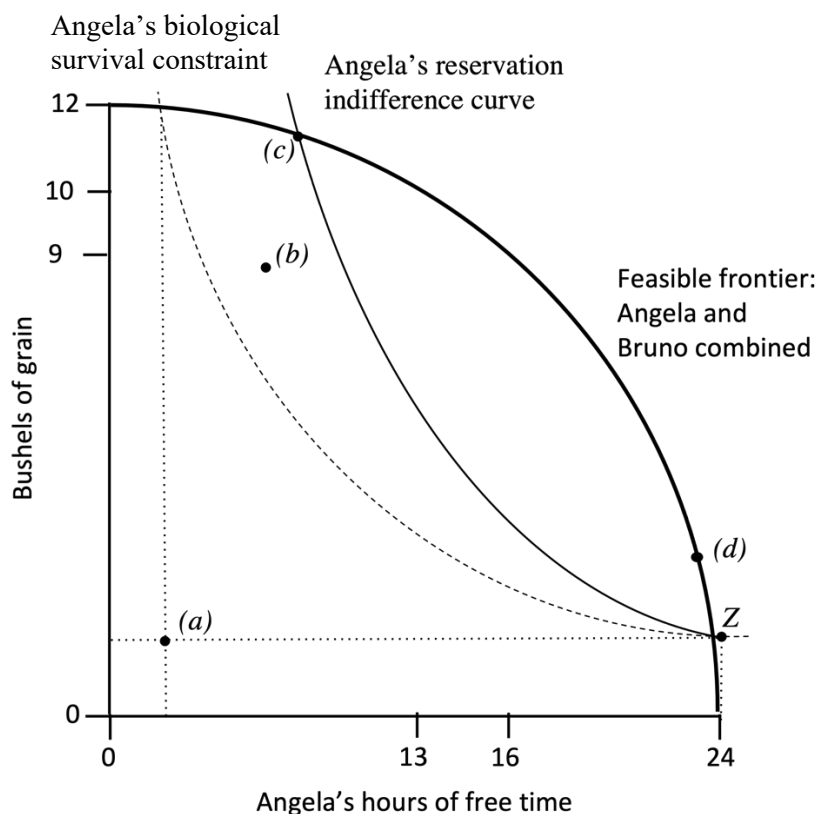
11. Carmen-Delta and Mbasakazi need to bring one type of snack to a party. Cupcakes are a favourite treat for both, but they generally prefer to have a mix of party food rather than eating cupcakes only. Carmen-Delta's second favourite snack is crisps, whereas Mbasakazi likes doughnuts more than crisps. The cellphone towers are not working so Carmen-Delta and Mbasakazi cannot talk to each other before the party. They therefore have to decide independently what snack to bring. The payoff matrix below represents their strategic interaction:

		Mbasakazi	
		Cupcakes	Doughnuts
Carmen-Delta	Cupcakes	2, 2	4, 5
	Crisps	5, 4	3, 3

Which one of the following statements is INCORRECT?

- a) Mbasakazi's best response is 'Doughnuts' if Carmen-Delta plays 'Cupcakes'.
 - b) This is a simultaneous-move game.
 - c) Neither player has a dominant strategy.
 - d) The Nash equilibrium of this game is (Crisps, Doughnuts).
12. In a problem involving exchange between two parties, the Pareto efficiency curve shows
- a) all exchanges that make both parties better off than they would be without an exchange.
 - b) the one exchange that makes both parties better off than they would be without an exchange.
 - c) all possible allocations of goods between both parties.
 - d) all possible efficient allocations between both parties.

13. Bruno is a landowner and Angela is a farmer who pays a share of her grain output to Bruno for the use of his land. Currently, if Angela does not work, she receives a ration of 2.5 bushels from the government. Of the four allocations (a), (b), (c) and (d) depicted in the figure, which allocation represents a Pareto improvement over Z?



- a) (a)
 - b) (b)
 - c) (c)
 - d) (d)
14. Tumi is hired by Stellenbosch University but does not earn an economic rent. What is the best explanation for Tumi's situation?
- a) Her reservation wage is less than the wage paid to her.
 - b) Her reservation wage is greater than the wage paid to her.
 - c) Her reservation wage is equal to the wage paid to her.
 - d) She is paid a wage that is lower than that paid to her co-workers.
15. As the hourly wage rises, the worker's best response function (with effort on the vertical axis and wages on the horizontal axis) becomes flatter. This statement is TRUE because
- a) as the level of effort approaches the maximum possible level, the disutility of marginal effort becomes greater.
 - b) the worker's goodwill towards the employer increases.
 - c) the marginal effect of effort on production is increasing.
 - d) The statement is FALSE.

16. Economies of scale exist when:
- a firm is large and diversified.
 - the average cost of producing a unit of output falls as output increases when all inputs are increased.
 - the average cost of producing a unit of output increases as output increases when all inputs are increased.
 - a firm is small and highly specialised.
17. An unregulated monopolist will produce the quantity at which the MR is equal to _____, but its P is equal to _____.
- WTP; MC
 - ATC; WTP.
 - MC; WTP.
 - WTA; WTP
18. Mary is a producer of a differentiated product. The demand curve for her product is given as follows:

$$Q = 160 - 4P$$

Mary's marginal revenue function is given by:

- $MR = 160 - \frac{1}{8}Q$
 - $MR = 40 - \frac{1}{8}Q^2$
 - $MR = 40 - \frac{1}{2}Q$
 - $MR = 160 - \frac{1}{4}Q^2$
19. Tabitha produces hats which she sells at R250 per hat. During the month of April, she sold 100 hats and made an economic profit of R10 000. If she produces the hats at a constant per unit cost, her average total cost of producing a hat is:
- R70.50
 - R170.50
 - R250.00
 - R150.00

20. The market for bread has many sellers and buyers, all of whom are price takers. Firms sell identical products. The price of wheat, one of the inputs in the production of bread, increases. Which one of the following statements is CORRECT?
- The market supply curve will shift to the left and the market demand curve will shift to the right. As a result, the equilibrium price of bread will increase.
 - The market supply curve will shift to the left, the equilibrium price of bread will increase, and the equilibrium quantity will decrease.
 - The market supply curve will shift to the right, the equilibrium price of bread will decrease, and the equilibrium quantity will increase.
 - The market demand curve will shift to the left, the equilibrium price of bread will decrease, and the equilibrium quantity will decrease.
21. Compared to a competitive industry, a monopolist:
- creates a deadweight loss and decreases economic profit.
 - creates a deadweight loss and decreases consumer surplus.
 - produces more output.
 - is more efficient because there is no unnecessary competition.
22. In December 2019 the price elasticity of demand for a firm's product is equal to 1. In July 2020 the price elasticity of demand is 1.5. This means that in July 2020
- the firm has less market power to increase prices since demand is now price-elastic.
 - the firm has more market power to increase prices since demand is now price-elastic.
 - the firm has less market power to increase prices since demand is now price-inelastic.
 - the firm has more market power to increase prices since demand is price-inelastic.
23. Every day in the Neelsie, 4 950 cups of coffee are sold at a price of R18 each. The price elasticity of demand for coffee is 1.3. If the price were to be increased to R22 per cup, what would total revenue per day be?
- R 25 520
 - R 77 440
 - R 108 900
 - R 141 570
24. Suppose that the market marginal private cost of producing a packet of OreosTM is $P = 100 + \frac{4}{75}Q$ and that the market demand curve for OreosTM is given by $P = 200 - \frac{1}{150}Q$. Further, suppose that the production of OreosTM emits pollutants that have a marginal external cost of $\frac{1}{150}Q$ (note: prices are in Rands). The optimal Pigouvian tax on a packet of OreosTM produced is _____ and tax revenue of _____ is generated.
- R11.11; R18 522
 - R10; R15 000
 - R12.50; R23 438
 - R12; R28 800

25. Suppose a firm imposes a negative externality on party B. The firm and party B could negotiate privately in order to reach an equilibrium that is socially optimal; however, Coasean bargaining may not always be practical to implement. Which of the following alternative solutions could result in a socially optimal outcome?
- I. The government could impose a regulation that caps the final output produced by the firm to a level that results in an efficient market outcome.
 - II. A Pigouvian tax can be imposed on the firm, where the tax is equal to the marginal social cost.
 - III. The legislator could enact a law that requires the firm to compensate party B for all damage caused.
- a) II only
 - b) I and II
 - c) I, II and III
 - d) I and III

SECTION C
LONG QUESTIONS
[60 marks]

QUESTION 1

Jack Sparrow is the captain of a ship called *The Flying Dutchman*. Another pirate, Barbossa, is the captain of a ship called *The Black Pearl*. Jack Sparrow would prefer to be captain of *The Black Pearl* and Barbossa would prefer to sail on *The Flying Dutchman*. The two pirates have the opportunity to trade ships. However, if one pirate chooses to trade and the other decides to keep his ship, then the second pirate (who does not trade) keeps both ships. The payoff matrix below represents their strategic interaction:

		Captain Jack Sparrow	
		Keep	Trade
Captain Barbossa	Keep	10, 10	13, 9
	Trade	9, 13	12, 12

1.1 Is (Trade, Trade) a Nash equilibrium? Why or why not? [2]

1.2 Explain whether the following statement is true or false: [2]

The pursuit of self-interest without regard for others always leads to social dilemmas.

QUESTION 2

Ruby works at the firm *Rossi Fisheries* and earns an hourly wage of R150. The tax rate she has to pay is equivalent to 5% of her gross hourly wage. At an effort level of 0.75, her displeasure of having to work with fish is equivalent to a cost of R25 per hour of work. If Ruby loses her job it will take her 30 days to find new employment. Ruby works 8 hour shifts per day.

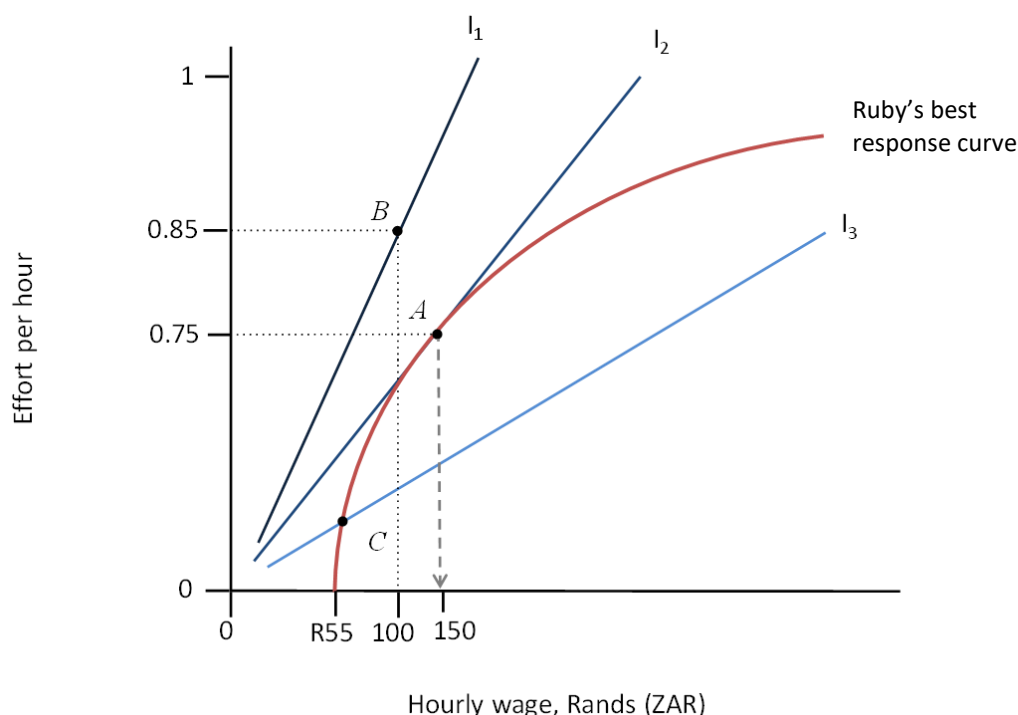
2.1 Using the information above, calculate Ruby's total employment rent. [3]

2.2 Ruby diligently contributed to the unemployment insurance fund (UIF) and can receive the equivalent of R55 per hour of unemployment benefits for 30 days. With this new information, what would her total employment rent now be equal to? [1]

2.3 Name four factors that can decrease Ruby's reservation wage. [2]

QUESTION 3

Study the figure below and answer the questions regarding the choice the firm *Rossi Fisheries* must make regarding Ruby's wage, taking into consideration Ruby's best response curve.



- 3.1 Which one of the isocost curves I_1 , I_2 and I_3 represents the firm's highest cost of effort?
Explain why. [2]
- 3.2 Would it be a feasible option for the firm to pay Ruby R55 per hour? Justify your answer. [1]
- 3.3 Which point (A, B or C) would be the most preferred for *Rossi Fisheries*, and why? [1]
- 3.4 At which point would the firm minimise its cost and maximise its profits? [1]
- 3.5 What hourly wage should the firm pay Ruby? [1]

QUESTION 4

A monopoly faces the following total cost and demand functions:

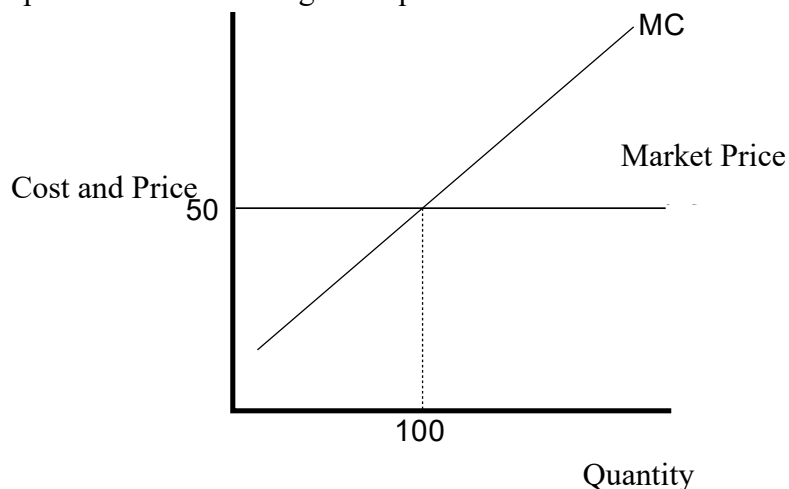
$$TC = 100 + 3Q + 0.5Q^2$$

$$Q = 240 - 4P$$

- 4.1 Write down a function for the firm's marginal cost (MC). [1]
- 4.2 Write down a function for the firm's marginal revenue (MR). [2]
- 4.3 Calculate the firm's profit maximising price and quantity. [4]
- 4.4 Calculate the firm's profit margin, $P - MC$. [1]
- 4.5 Calculate the firm's total economic profit. [2]

QUESTION 5

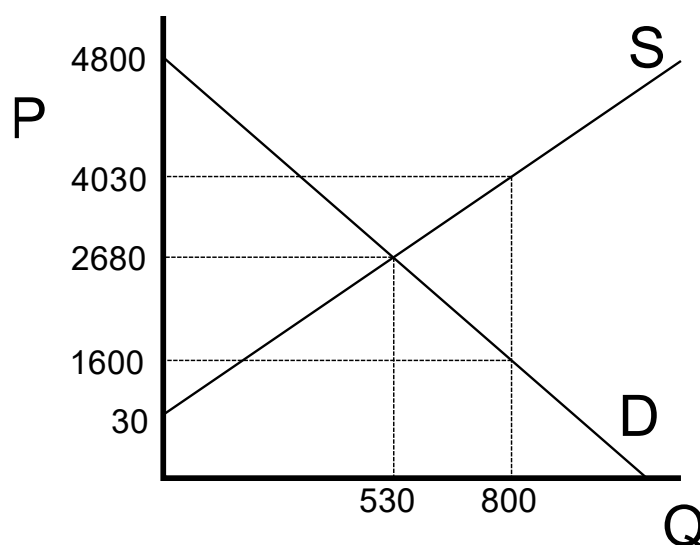
The graph below shows the marginal cost of production and the market price for a firm in a perfectly competitive market in long-run equilibrium.



- 5.1 Define the concept marginal revenue. [1]
- 5.2 What is this firm's marginal revenue? Briefly explain your answer. [2]
- 5.3 What is this firm's average cost? Briefly explain your answer. [1]
- 5.4 Explain why the firm earns the maximum possible profit when it chooses to supply a quantity of 100. [3]
- 5.5 If this firm (and only this firm) were to discover a new technology that reduces its marginal cost, what would the short-run effect on the firm's quantity supplied be? Briefly explain (no graph needed). [2]
- 5.6 Suppose that the new technology eventually becomes available to other firms in the market as well. What will the effect on equilibrium market price and quantity be in the short run? How will this firm's quantity supplied adjust in response to these developments? Graphs are not required. [3]

QUESTION 6

The graph below (not drawn according to scale) indicates the demand and supply curves in a competitive market.



6.1 Calculate the total surplus in this market at the equilibrium quantity. [2]

6.2 Calculate the deadweight loss and the total surplus in this market if 800 units are produced and consumed. [3]

QUESTION 7

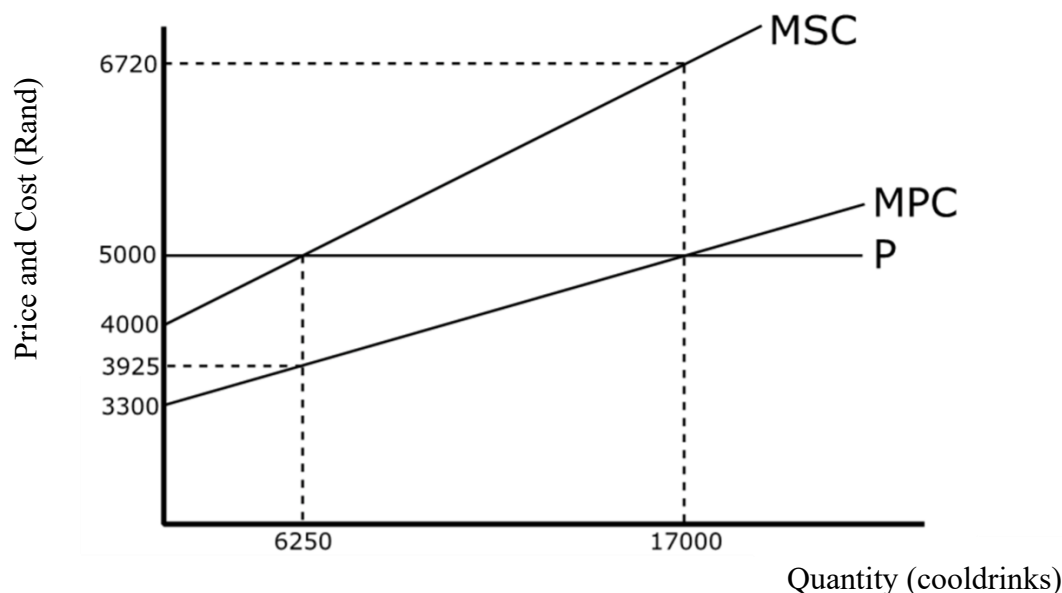
Explain, with the aid of a graph, how the burden of a tax imposed on sellers is divided between buyers and sellers in a market with demand that is relatively price-inelastic (compared to the price elasticity of supply). [7]

QUESTION 8

Discuss two methods, other than advertising, that firms can use to increase their market power. For each, make clear why this will give the firm market power. [4]

QUESTION 9

A cooldrink factory, *Fizz*, produces carbon dioxide as a by-product in the manufacturing process, which damages a nearby strawberry farm. The graph below (not drawn to scale) shows the demand, MPC and MSC of cooldrinks. The cooldrinks market is competitive.



- 9.1 What is the equilibrium quantity of cooldrinks in an unregulated market? [1]
- 9.2 What is the Pareto-efficient quantity of cooldrinks? [1]
- 9.3 Suppose that the strawberry farmer is bearing the cost of the pollution and she complains to her local municipality. Consequently, the government decides to regulate the cooldrink market, and they impose a tax on the cooldrinks produced by *Fizz*. What is the optimal Pigouvian tax per can of cooldrink that the government should implement? [1]
- 9.4 Now suppose that there is no tax, and the strawberry farmer negotiates (bargains) directly with the cooldrink manufacturer. The farmer offers to pay *Fizz* to produce fewer cooldrinks. What is the minimum acceptable offer per unit from the farmer to *Fizz* to ensure a Pareto-efficient outcome? [2]
- 9.5 Briefly describe one obstacle to bargaining that may prevent the achievement of Pareto efficiency. [1]

**UNIVERSITEIT VAN STELLENBOSCH
DEPARTEMENT EKONOMIE**

EKONOMIE 114

HOOFASSESSERING A2: 2 JUNIE 2022

Tyd: 3 ure

Punte: 100

EKSAMINATORE: Dr K Malindi Me L Boltman
 Prof S du Plessis Me C Pengelly
 Dr D Shepherd Me J Terblanche
 Mnr MP van Schoor

INTERNE MODERATOR: Dr E Moses

LET WEL: SLEGS NIE-PROGRAMMEERBARE SAKREKENAARS WORD TOEGELAAT.

BEANTWOORD **AFDELING A** VRAE OP DIE **ANTWOORDBLAD** WAT VERSKAF WORD. TOON U KEUSE DEUR 'N **KRUISIE** LANGS DIE NOMMER VAN DIE STELLING TE MAAK.

BEANTWOORD OOK **AFDELING B** VRAE OP DIE **ANTWOORDBLAD** WAT VERSKAF WORD. TOON U KEUSE DEUR 'N **KRUISIE** LANGS DIE NOMMER VAN DIE STELLING TE MAAK.

SLEUTEL:

AP = Gemiddelde produk
AFC = Gemiddelde vaste koste
AVC = Gemiddelde veranderlike koste
AC = Gemiddelde totale koste
BL = Begrotingslyn
BBP = Bruto Binnelandse Produk
IC = Indifferensie kurwe
K = Kapitaal
FF = Haalbare kurwe
MC = Marginale koste
MP = Marginale produk
MPB = Marginale private voordeel
MPC = Marginale private koste
MRS = Marginale koers van substitusie

MRT = Marginale koers van transformasie
MSB = Marginale sosiale voordeel
MSC = Marginale sosiale koste
P = Prys
PF = Produksiemoontlikheidskurwe
Q = Hoeveelheid
Q_d = Hoeveelheid gevra
Q_s = Hoeveelheid aangebied
TC = Totale koste
TP = Totale produk
w = Loon
WTA = Bereidheid om te aanvaar
WTP = Bereidheid om te betaal
Y = Inkome

AFDELING A

WAAR/ONWAAR STELLINGS

[10 punte: 1 punt vir 'n korrekte antwoord en minus ½ vir 'n verkeerde antwoord]

1. In 'n spel met twee spelers, waar Speler 1 drie moontlike strategieë het waaruit sy kan kies en Speler 2 drie moontlike strategieë het waaruit hy kan kies, is daar nege moontlike uitkomst.
2. Die onderstaande spel is 'n voorbeeld van die gevangenes se dilemma-spel.

		Sam	
		Dra masker	Dra nie masker
Jemima	Dra masker	5, 5	2, 6
	Dra nie masker	6, 2	3, 3

- a) Waar
 - b) Onwaar
3. Chani is 'n grondeienaar en Paul is 'n speseryboer wat 'n deel van sy produksie aan haar betaal vir die gebruik van haar grond. Nuwe tegnologie wat Paul se produktiwiteit verhoog, sal die oppervlakte van die ekonomies haalbare stel vergroot.
 4. 'n Toedeling sal wesenlik (substantief) regverdig wees vir alle deelnemers aan 'n strategiese interaksie, as dit Pareto-doeltreffend is.
 5. Wanneer werkers hul reserweloan in die arbeidsdissipline model betaal word, het werkgewers geen mag oor hulle nie.
 6. Die prinsipaal-agent-probleem van eienaarskap versus beheer van die firma ontstaan wanneer eienaars en bestuurders doelwitte nastreef wat verskil van dié wat die regering wil hê hulle moet nastreef.
- a) Waar
 - b) Onwaar

7. Die vraag na uitgevoerde Russiese aardgas is prys-onelasties. Indien Rusland sy aanbod van gasuitoere met 5 persent verminder, sal die prys van aardgas met meer as 5 persent styg.
- a) Waar
 - b) Onwaar
8. Daar is twee verkopers van "soft serve" roomys in Funtown. Op 'n dag besluit roomysverkoper A om die kwaliteit van haar produk te verbeter deur sjokoladestrooisels by haar produk te voeg. Dit lok aanvanklik sommige klante weg van roomysverkoper B se kiosk, maar B kry dit gou reg om al die verlore klante terug te kry deur 'n prysverlaging aan te bied. Op die ou end is B se hoeveelheid roomyse wat verkoop word dieselfde as voorheen. Uit hierdie veranderinge kan ons aflei dat die vraagkurwe vir B se roomyse eers linkswaarts en toe regswaarts verskuif het.
- a) Waar
 - b) Onwaar
9. Indien daar negatiewe eksternaliteite in 'n ekonomie met winsmaksimerende ondernemings en mededingende markte (geen owerheidsinmenging) bestaan, sal daar oorproduksie wees en $MSB > MSC$.
- a) Waar
 - b) Onwaar
10. Wanneer markte hulpbronne op 'n Pareto-ondoeltreffende wyse toedeel, staan dit bekend as markmislukking.
- a) Waar
 - b) Onwaar

AFDELING B
VEELVULDIGE-KEUSE VRAE

(30 punte: 2 punte vir 'n korrekte antwoord en minus ¼ vir 'n verkeerde antwoord)

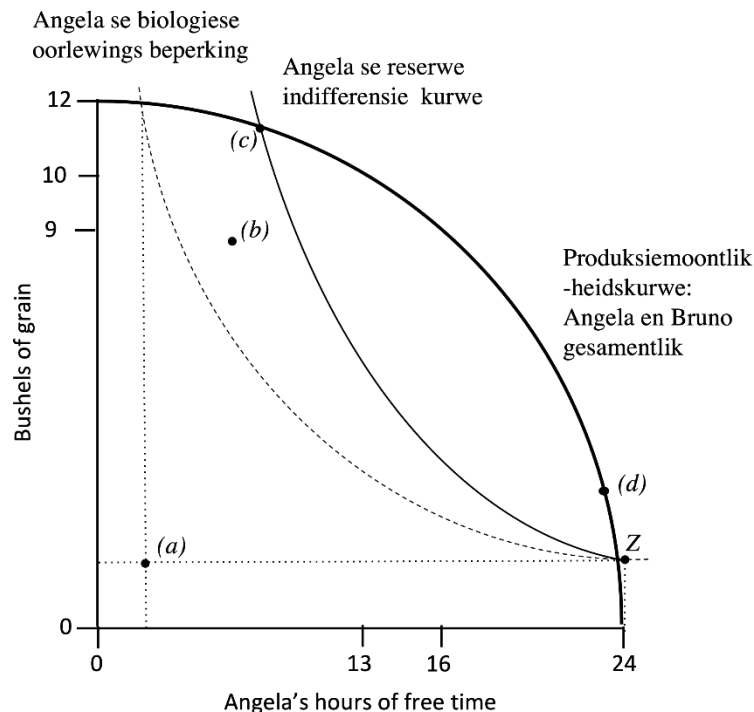
11. Carmen-Delta en Mbasakazi moet elkeen een tipe peuselhappie na 'n partytjie toe neem. Kolwyntjies is beide van hulle se gunsteling peuselhappie, maar hulle verkies gewoonlik om 'n mengsel van partyjiekos te eet, eerder as om slegs kolwyntjies te eet. Carmen-Delta se tweede gunsteling peuselhappie is skyfies terwyl Mbasakazi meer van oliebolle as skyfies hou. Die selfoontorings is buite werking en gevolglik kan Carmen-Delta en Mbasakazi nie voor die partytjie met mekaar praat nie en dus moet hulle onafhanklik besluit watter peuselhappie om te bring. Die uitbetalingsmatriks hieronder verteenwoordig hulle strategiese interaksie:

		Mbasakazi	
		Kolwyntjies	Oliebolle
Carmen-Delta	Kolwyntjies	2, 2	4, 5
	Skyfies	5, 4	3, 3

Watter een van die volgende stellings is VERKEERD?

- Mbasakazi se beste respons is 'Oliebolle' indien Carmen-Delta 'Kolwyntjies' speel.
 - Dit is 'n gelyktydige-beweging spel.
 - Nie een van die spelers het 'n dominante strategie nie.
 - Die Nash ewewig van hierdie spel is (Skyfies, Oliebolle).
12. In 'n probleem wat uitruiling tussen twee partye behels, wys die Pareto-doeltreffendheidskurwe
- alle uitruilings wat beide partye beter daaraan toe maak as wat hulle sou wees sonder uitruiling.
 - die een uitruiling wat beide partye beter daaraan toe maak as wat hulle sou wees sonder uitruiling.
 - alle moontlike toedelings van goedere tussen beide partye.
 - alle moontlike doeltreffende toedelings tussen beide partye.

13. Bruno is 'n grondeienaar en Angela is 'n boer wat 'n deel van haar graanproduksie aan Bruno betaal vir die gebruik van sy grond. Tans, as Angela nie werk nie, ontvang sy 'n rantsoen van 2.5 gerwe koring van die regering. Van die vier toedelings (a), (b), (c), en (d) wat in die figuur uitgebeeld word, watter toedeling verteenwoordig 'n Pareto verbetering oor Z?



- a) (a)
b) (b)
c) (c)
d) (d)
15. Tumi word deur die Universiteit van Stellenbosch aangestel, maar verdien nie 'n ekonomiese opbrengs ("rent") nie. Wat is die beste verduideliking vir Tumi se situasie?
- a) Haar reserweloan is minder as die loon wat sy betaal word.
b) Haar reserweloan is meer as die loon wat sy betaal word.
c) Haar reserweloan is gelyk aan die loon wat sy betaal word.
d) Sy word 'n loon betaal wat laer is as dié wat aan haar medewerkers betaal word.
15. Soos die uurlikse loon styg, word die werker se beste reaksiekurwe (met inspanning (werksywer) op die vertikale as en lone op die horisontale as) platter. Hierdie stelling is WAAR omdat
- a) namate die vlak van inspanning die maksimum moontlike vlak nader, word die negatiewe nut van marginale inspanning groter.
b) die werker se welwillendheid teenoor die werkgever toeneem.
c) die marginale effek van inspanning op produksie toeneem.
d) Die stelling is ONWAAR.

16. Skaalvoordele bestaan wanneer:
- 'n firma groot en gediversifiseerd is.
 - die gemiddelde koste om een eenheid van uitset te produseer daal namate die uitset toeneem wanneer alle insette vermeerder word.
 - die gemiddelde koste om een eenheid van uitset te produseer toeneem namate die uitset toeneem wanneer alle insette vermeerder word.
 - 'n firma klein en hoogs gespesialiseerd is.
17. 'n Ongereguleerde monopolis sal die hoeveelheid produseer waar die MR gelyk is aan _____, maar die P daarvan is gelyk aan _____.
- WTP; MC
 - ATC; WTP
 - MC; WTP
 - WTA; WTP
18. Marie produseer 'n gedifferensieerde produk. Die vraagkurwe vir haar produk is soos volg:
- $$Q = 160 - 4P$$
- Marie se marginale inkome funksie word gegee deur:
- $MR = 160 - \frac{1}{8}Q$
 - $MR = 40 - \frac{1}{8}Q^2$
 - $MR = 40 - \frac{1}{2}Q$
 - $MR = 160 - \frac{1}{4}Q^2$
19. Tabitha vervaardig hoede wat sy verkoop teen R250 elk. Gedurende die maand van April het sy 100 hoede verkoop en 'n ekonomiese wins van R10 000 gemaak. Indien sy die hoede teen 'n konstante koste per eenheid produseer, is haar gemiddelde totale koste om 'n hoed te produseer:
- R70.50
 - R170.50
 - R250.00
 - R150.00
20. Die mark vir brood het baie verkopers en kopers. Almal is prysnemers en die ondernemings verkoop identiese produkte. Die prys van graan, een van die insette in die produksie van brood, styg. Watter een van die volgende stellings is KORREK?
- Die markaanbodkurwe sal na links skuif en die markvraagkurwe sal na regs skuif. Die ewewigsprys van brood sal gevolglik toeneem.
 - Die markaanbodkurwe sal na links skuif, die ewewigsprys van brood sal toeneem en die ewewigshoeveelheid sal afneem.
 - Die markaanbodkurwe sal na regs skuif, die ewewigsprys van brood sal afneem en die ewewigshoeveelheid sal toeneem.

- d) Die markvraagkurwe sal na links skuif, die ewewigsprys van brood sal afneem en die ewewigshoeveelheid sal afneem.
21. In vergelyking met 'n mededingende industrie, sal 'n monopolis:
- 'n dooiegewigsverlies veroorsaak en ekonomiese wins verlaag.
 - 'n dooiegewigsverlies veroorsaak en verbruikersurplus verlaag.
 - Meer uitset produseer.
 - meer doeltreffend wees omdat daar geen onnodige mededinging is nie.
22. In Desember 2019 is die pryselastisiteit van vraag na 'n firma se produk gelyk aan 1. In Julie 2020 is die pryselastisiteit van vraag 1.5. Dit beteken dat in Julie 2020
- die firma minder markmag het om die prys te verhoog aangesien vraag nou prys-elasties is.
 - die firma meer markmag het om die prys te verhoog aangesien vraag nou prys-elasties is.
 - die firma minder markmag het om die prys te verhoog nie aangesien vraag nou prys-onelasties is.
 - die firma meer markmag het om die prys te verhoog aangesien vraag nou prys-onelasties is.
23. Elke dag in die Neelsie word 4 950 koppies koffie teen 'n prys van R18 elk verkoop. Die pryselastisiteit van vraag na koffie is 1.3. As die prys tot R22 per koppie verhoog sou word, wat sou die totale inkome per dag wees?
- R 25 520
 - R 77 440
 - R 108 900
 - R 141 570
24. Veronderstel dat die mark marginale private koste om 'n pakkie OreosTM te produseer $P = 100 + \frac{4}{75}Q$ is, en dat die mark vraagkurwe vir OreosTM deur $P = 200 - \frac{1}{150}Q$ voorgestel word. Veronderstel verder dat die produksie van OreosTM besoedeling vrystel wat 'n marginale eksterne koste van $\frac{1}{150}Q$ het (let wel: pryse is in Rande). Die optimale Pigouviaanse belasting op die produksie van 'n pakkie OreosTM is _____, en belastinginkomste van _____ word gegenereer.
- R11.11; R18 522
 - R10; R15 000
 - R12.50; R23 438
 - R12; R28 800
25. Gestel 'n firma imponeer 'n negatiewe eksternaliteit op party B. Die firma en party B kan privaat onderhandel om 'n ewewig te bereik wat sosiaal optimaal is; Coasean-bedinging is egter nie altyd prakties om te implementeer nie. Watter van die volgende alternatiewe oplossings kan tot 'n sosiaal optimale uitkoms lei?

- I. Die regering kan 'n regulasie instel wat die finale uitset wat deur die firma geproduseer word, beperk tot 'n vlak wat 'n doeltreffende markuitkoms tot gevolg het.
 - II. 'n Pigouvianse belasting kan op die firma gehef word, waar die belasting gelyk is aan die marginale sosiale koste.
 - III Die wetgewer kan 'n wet uitvaardig wat vereis dat die firma party B moet vergoed vir enige skade veroorsaak.
- a) Slegs II
 - b) I en II
 - c) I, II en III
 - d) I en III

AFDELING C
LANG VRAE
[60 punte]

VRAAG 1

Jack Sparrow is die kaptein van 'n skip genaamd "*The Flying Dutchman*". 'n Ander seerower, Barbossa, is die kaptein van 'n skip genaamd "*The Black Pearl*". Jack Sparrow sal verkies om die kaptein van "*The Black Pearl*" te wees en Barbossa sal verkies om op "*The Flying Dutchman*" te vaar. Die twee seerowers het die opsie om skepe te ruil. Indien een seerower egter kies om te ruil en die ander een besluit om sy skip te behou, sal die tweede seerower (wat nie geruil het nie) beide skepe hou. Die onderstaande betalingsmatriks verteenwoordig hul strategiese interaksie:

		Kaptein Jack Sparrow	
		Hou	Ruil
Kaptein Barbossa	Hou	10, 10	13, 9
	Ruil	9, 13	12, 12

1.1 Is (Ruil, Ruil) 'n Nash ewewig? Hoekom of hoekom nie? [2]

1.2 Verduidelik of die volgende stelling waar of onwaar is: [2]

Die strewe na selfbelang sonder die inagneming van ander lei altyd tot sosiale dilemmas.

VRAAG 2

Ruby werk by die firma *Rossi Visserye* en verdien 'n uurlikse loon van R150. Die belastingkoers wat sy moet betaal is gelykstaande aan 5% van haar bruto uurlikse loon. Op 'n inspanningsvlak van 0.75 is haar misnoeë om met vis te moet werk gelykstaande aan 'n koste van R25 per uur se werk. Indien Ruby haar werk verloor, sal dit haar 30 dae neem om nuwe werk te kry. Ruby werk 8-uur skofte per dag.

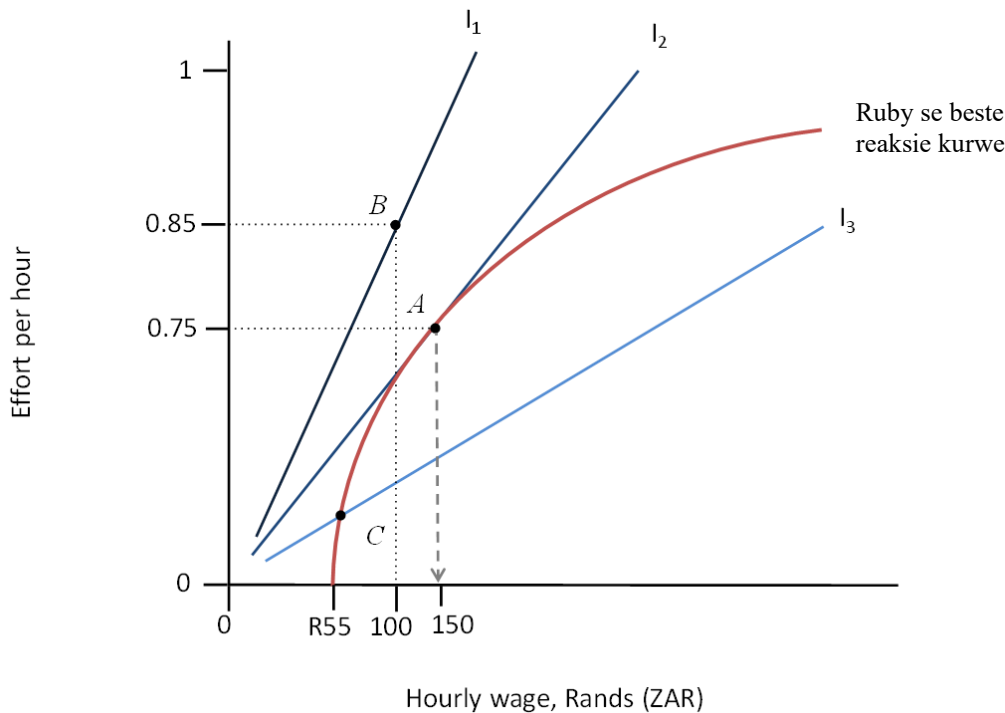
2.1 Gebruik die inligting hierbo om Ruby se totale indiensnemingsopbrengs ("rent") te bereken. [3]

2.2 Ruby het ywerig tot die werkloosheidsversekeringsfonds (UIF) bygedra en kan die ekwivalent van R55 per uur se werkloosheidsvoordele vir 30 dae ontvang. Met hierdie nuwe inligting, wat sou haar totale indiensnemingsopbrengs ("rent") nou wees? [1]

2.3 Noem vier faktore wat Ruby se reserweloan kan verlaag. [2]

VRAAG 3

Bestudeer die onderstaande figuur om die vrae te beantwoord oor die keuse wat die firma *Rossi Visserie* moet maak ten opsigte van Ruby se loon, met inagneming van Ruby se beste responskurwe.



- 3.1 Watter een van die isokostekurwes I_1 , I_2 en I_3 verteenwoordig die firma se hoogste koste vir inspanning? Verduidelik waarom. [2]
- 3.2 Sou dit 'n haalbare opsie wees vir die firma om Ruby R55 per uur te betaal? Motiveer jou antwoord. [1]
- 3.3 Watter punt (A, B of C) sal die mees verkose vir *Rossi Visserie* wees, en waarom? [1]
- 3.4 Op watter punt sal die onderneming sy koste tot die minimum beperk en sy wins maksimeer? [1]
- 3.5 Watter uurlikse loon moet die firma Ruby betaal? [1]

VRAAG 4

'n Monopolie het die volgende totale koste en vraagfunksies:

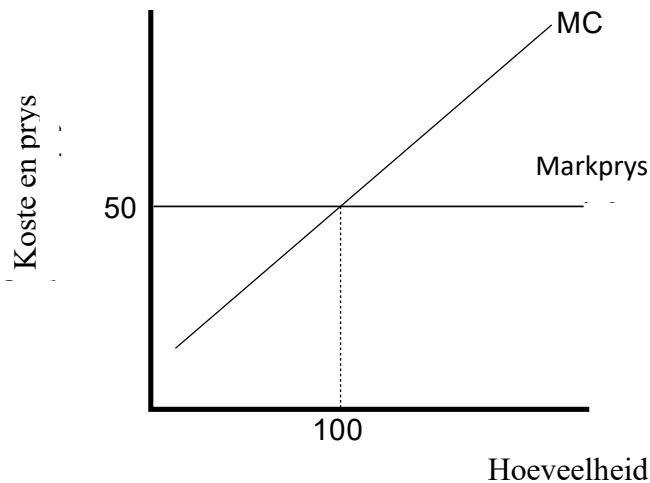
$$TC = 100 + 3Q + 0.5Q^2$$

$$Q = 240 - 4P$$

- 4.1 Skryf 'n funksie vir die firma se marginale koste (MC) neer. [1]
- 4.2 Skryf 'n funksie vir die firma se marginale inkome (MR) neer. [2]
- 4.3 Bereken die firma se winsmaksimerende prys en hoeveelheid. [4]
- 4.4 Bereken die firma se winsgrens, $P - MC$. [1]
- 4.5 Bereken die firma se totale ekonomiese wins. [2]

VRAAG 5

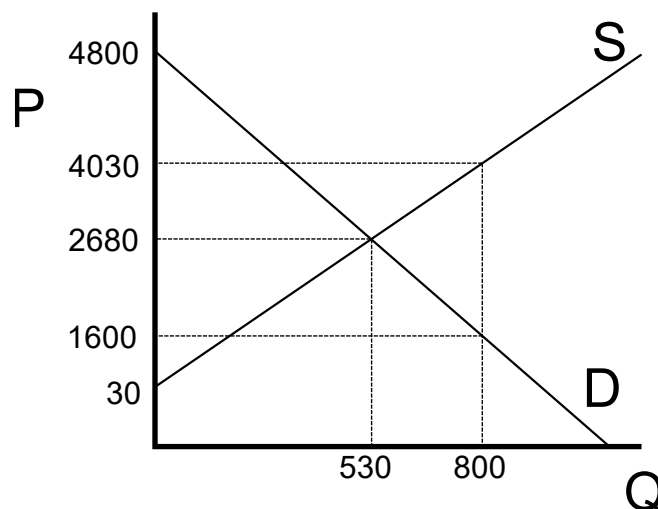
Die grafiek hieronder toon die marginale koste van produksie en markprys vir 'n firma in 'n volmaak mededingende mark in langtermyn-ewewig.



- 5.1 Definieer die begrip marginale inkome. [1]
- 5.2 Wat is hierdie firma se marginale inkome? Verduidelik kortliks jou antwoord. [2]
- 5.3 Wat is hierdie firma se gemiddelde koste? Verduidelik kortliks jou antwoord. [1]
- 5.4 Verduidelik waarom die firma die maksimum moontlike wins verdien wanneer hy kies om 'n hoeveelheid van 100 eenhede aan te bied. [3]
- 5.5 Indien hierdie firma (en slegs hierdie firma) 'n nuwe tegnologie ontdek wat sy marginale koste verminder, wat sal die effek op die firma se hoeveelheid aangebied wees? Verduidelik kortliks (geen grafiek nodig nie). [2]
- 5.6 Veronderstel dat die nuwe tegnologie uiteindelik ook vir ander firmas in die mark beskikbaar word. Wat sal die effek op ewewigsmarkprys en -hoeveelheid in die kort termyn wees? Hoe sal hierdie firma sy hoeveelheid aangebied aanpas as gevolg van hierdie verwickelinge? Grafieke is nie nodig nie. [3]

VRAAG 6

Die onderstaande grafiek (nie volgens skaal geteken nie) dui die vraag- en aanbodkurwes in 'n mededingende mark aan.



- 6.1 Bereken die totale surplus in hierdie mark by die ewewigshoeveelheid. [2]
- 6.2 Bereken die dooiegewigsverlies en die totale surplus in hierdie mark indien 800 eenhede geproduseer en verbruik word. [3]

VRAAG 7

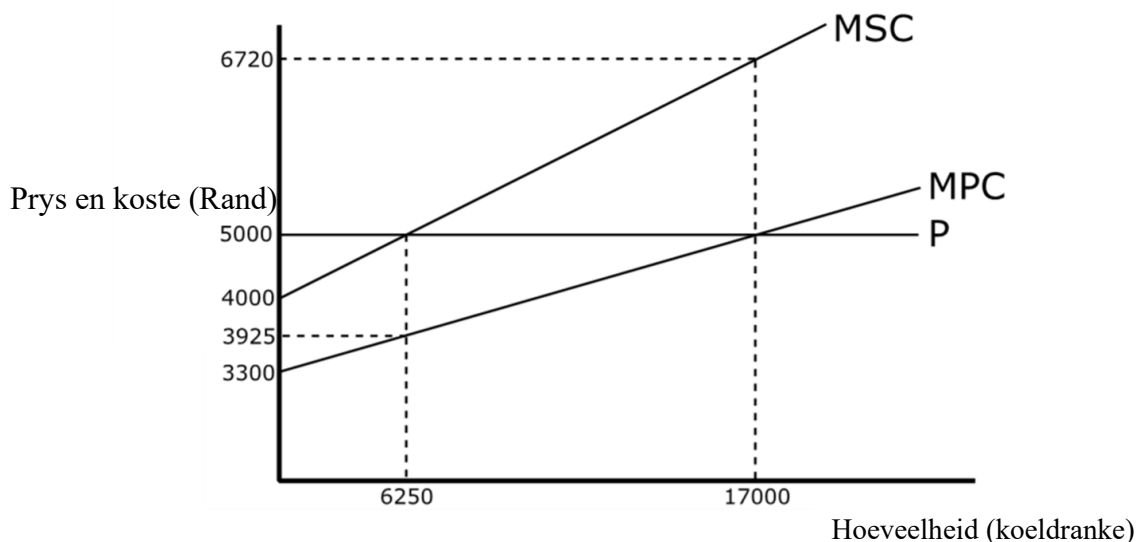
Verduidelik, met behulp van 'n grafiek, hoe die las van 'n belasting gehef op verkopers verdeel word tussen kopers en verkopers in 'n mark met 'n vraag wat relatief prys-onelasties is (vergeleke met die pryselastisiteit van aanbod). [7]

VRAAG 8

Bespreek twee maniere, buiten advertensies, wat firmas kan gebruik om hul markkrag te verhoog. Maak dit in elke geval duidelik waarom dit tot meer markkrag vir die firma sal lei. [4]

VRAAG 9

'n Koeldrankfabriek, Fizz, produseer koolstofdiksied, 'n by-produk in die vervaardigingsproses, wat 'n nabye arbeiplaas beskadig. Die grafiek hieronder (nie op skaal geteken nie) wys die vraag, MPC en MSC van koeldranke. Die mark vir koeldranke is mededingend.



- 9.1 Wat is die ewewigshoeveelheid van koeldranke in 'n ongereguleerde mark? [1]
- 9.2 Wat is die Pareto-doeltreffende hoeveelheid koeldranke? [1]
- 9.3 Veronderstel dat die arbeiboer die koste van die besoedeling dra en dat sy by haar plaaslike munisipaliteit kla. Gevolglik besluit die regering om die koeldrankmark te reguleer en hulle stel 'n belastingkoers op die koeldranke wat deur Fizz geproduseer word, in. Wat is die optimale Pigouvianse belasting wat die regering op 'n blikkie koeldrank moet implementeer? [1]
- 9.4 Veronderstel nou dat daar geen belasting is nie en dat die arbeiboer direk met die koeldrankvervaardiger onderhandel (beding). Die boer bied aan om Fizz te betaal om minder koeldranke te produseer. Wat is die minimum aanvaarbare aanbod per eenheid van die boer aan Fizz sodat daar 'n Pareto-doeltreffende uitkoms kan wees? [2]
- 9.5 Verduidelik kortliks een struikelblok tot bedinging wat die bereiking van Pareto-doeltreffendheid kan verhoed. [1]